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Warnock, A. C. C.

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Sound Transmission Through Concrete Block Walls

by A.C.C. Warnock

Dr. A.C.C. Warnock is a senior research officer in the Acoustics Laboratory of the Institute for Research in Construction.

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Sound insulation requirements in the 1990 edition of the National Building Code of Canada are to be increased in response to a general demand for greater sound insulation in homes. Some jurisdictions in Canada are already asking for sound transmission class (STC) ratings of 55 or better. There is also an increasing recognition that sound at low frequencies, below the limit normally measured in tests, is a major cause of complaint in buildings. While voices are not strong low-frequency sources, modern stereos can be. Mechanical equipment next to living areas are frequent sources of low-frequency noise and cause complaints. For economical control of noise in buildings, knowledge of the factors that control sound transmission through wall systems is important.

A recent series of measurements at the Institute for Research in Construction on behalf of the Ontario Concrete Block Association (OCBA) looked at methods of obtaining high sound transmission class ratings and good low-frequency sound attenuation with block walls. Despite the fact that walls of concrete blocks and other types of walls have been used for many years, there were several reasons for new transmission loss measurements:

- There are discrepancies among data presented in literature for blocks with and without finished surfaces. Some of these can be ascribed to improvements in measurement techniques that render old data obsolete, others to differences in construction and laboratory facilities, and in the physical structure of the blocks from different manufacturers.
- There is a lack of test data on block wall systems with very high STC ratings. This despite the fact that it is relatively simple to get high ratings and that block walls are often the system of choice for reducing high levels of noise from machine rooms.
- Only a few laboratories have begun to collect sound transmission information at low frequencies. Most of the transmission loss information in published data does not extend below 125 Hz, the lowest frequency required to calculate the STC.

The following text is a summary of information available in the literature and the major results of the OCBA study. A summary of the sound transmission properties of single-wythe blocks is also included. Some measurements on two-leaf block walls illustrate their potential high performance, while showing the installation problems that must be dealt with to ensure that the high performance is obtained. Test results from an extensive series of measurements on block walls with additional layers of drywall show that such systems can give excellent sound insulation. It is also easier to control construction errors with these types of walls.

Sound Insulation of a Single-Leaf Block Wall

For significant noise reduction between two rooms, the wall or floor separating them must transmit only a tiny fraction of the sound energy striking it. The ratio of the sound energy striking the wall to the transmitted sound energy, expressed in decibels (dB), is called the transmission loss (TL). The less sound energy transmitted, the higher the transmission loss. Figure 1 shows the transmission loss measured for the 190-mm (nom. 8") thick concrete-block wall used in most of the tests reported here. The curve shows typical behaviour for blocks; transmission loss is lower at low frequencies and increases about 6 dB each time the frequency doubles.

Sound transmission class (STC) is a single-number rating that summarizes transmission loss data; it is obtained by fitting a standard reference contour to the data. The fitted STC 50 contour is also shown in Figure 1. Transmission loss values that fall below the contour determine the STC value. No single transmission loss value may lie more than 8 dB below the contour.

Figure 2 shows results for some other block walls measured at IRC. The fluctuations in the transmission loss contours at low frequencies are due to the stiffness and other physical characteristics of the block. Thicker blocks do not always perform better than thinner ones at all frequencies.

An increase in transmission loss is expected with increasing mass, because the heavier the wall, the less it vibrates in response to sound waves, hence the less sound energy it radiates on the other side. Figure 3 shows measured sound transmission class ratings for single-wythe block walls from a number of sources in the literature. The scatter is considerable-confirming that block weight is not the only factor that determines sound transmission class.

Table 1.STC ratings for standard hollow normal-weight and lightweight block walls sealed on at least one side

Nominal	lightweight		normal	
thickness (mm)	kg/block	STC	kg/block	STC
100	8	43	10	44
150	10	44	15	46
200	14	46	18	48
250	17	47	21	49
300	20	49	25	51

Previous work has shown that where the block is porous, sealing the surface can have a significant effect on the sound transmission; the more porous the block, the more sound will leak through the block structure and the greater will be the improvement due to sealing. Improvements of 3 to 5 STC points are not uncommon for some lightweight block walls after they have been sealed. For some lightweight blocks, the improvements can be even more; one 290-mm (nom. 12") lightweight block, tested at IRC, increased from STC 39 to STC 50 after painting. Conversely, some blocks do not leak significantly and show no improvement after sealing. Nevertheless, to ensure the maximum performance from a block or masonry wall that is not to have drywall attached to it, all the joints must be properly finished and it is safer to seal the wall completely on at least one side using plaster or coats of block sealer. Leaking walls were eliminated as far as possible from the plotted data shown in Figure 3, but some of the results that lie far from the regression line may still be from block walls where sound is leaking through unsealed surfaces.

Concrete blocks can vary widely in physical properties from one manufacturer to another. The combination of the effects of weight, stiffness, porosity, and the shape of normal hollow block is too complicated to allow accurate prediction of sound transmission loss. One has to fall back on empirical approaches and measurement. However, in the absence of measured data, the regression line from Figure 3 can be used to predict STC. Table 1 gives STC values extracted from the graph for normal hollow blocks that have been sealed on at least one side.

Improving Single-Leaf Block Walls

Figure 3 shows that using heavier block to get a greater STC rating leads to walls that are impractically heavy except in special circumstances; the maximum STC in the figure is 56 for a block weight of 36 kg (nom. 80 lb). Adding materials, such as sand or grout to the cores of the blocks, makes them perform like solid blocks; the increase in transmission loss due to the increase in weight can be estimated from Figure 3. Adding sound-absorbing materials is usually not very effective because the transmission is primarily through the structure of the block. Two-leaf masonry walls can also be used in critical applications, but require especially careful installation. It will be shown that high STC ratings are more easily obtained by adding layers of drywall mounted on studs or furring.

Two-Leaf Masonry Walls

Two-leaf masonry walls can, in principle, provide very high sound insulation. They appear to be ideal: two independent, heavy layers separated by an airspace. The improvement over a single-wythe

wall with the same total weight should be considerable. Against this one must weigh the practical difficulties associated with constructing two block wythes that are not solidly connected. In practice, there is always some transmission of energy along the floor and ceiling, along walls abutting the periphery of the cavity wall, and through other parts of the structure. Sound can travel along wire ties and seriously impair the sound insulation, unless they are sufficiently flexible. Physical breaks in the floor, ceiling, and abutting walls are needed to reduce transmission along these flanking paths. Even if construction breaks are included in the design, mortar droppings or some other error can bridge the gap and increase sound transmission. Figure 4 shows results for a two-leaf block wall formed from two 90-mm (nom. 4") wythes with glass-fibre in the cavity; the wythes were well isolated. Also shown are the results for a wall of heavier block, but with polystyrene insulation replacing the glass-fibre and with mortar droppings reducing the isolation between the wythes. Despite the heavier block, the faulty construction inside resulted in a much lower transmission loss. Such errors are usually concealed and impossible to fix after the wall is complete.

Table 2 gives some results for two-leaf cavity block walls tested in the IRC laboratory; most had both wythes structurally isolated from each other. This installation provided the least amount of structural vibration transmission that can be achieved in the laboratory and is better than could be achieved in normal field installations. In cavity walls, adding sound-absorbing material to the cavity usually produces increased sound transmission losses and an increased STC. It is interesting to see that, even in heavy constructions like these, there is still something to be gained by using effective sound absorbers in the cavity. On the other hand, foam-plastic insulation does not absorb sound well; it provides no increase in STC. The last wall in Table 2 was tested with both wythes supported on the same mounting frame; this allows sound energy to travel from one wythe to the other through the supporting structure. The STC dropped a further 7 points relative to the same wall with complete isolation.

Table 2.Summary of construction and STC ratings for two-leaf block cavity walls; all except the last had the two wythes structurally isolated

First wythe	Cavity	Second wythe	STC
190-mm blocks 16-mm drywall screwed to blocks	65-mm glass-fibre panels, 90-mm air space, isolated	90-mm split-rib blocks	79
90-mm blocks 16-mm drywall screwed to blocks isolated	65-mm glass-fibre panels, 60-mm air space, isolated	90-mm split-rib blocks	77
90-mm blocks 16-mm drywall screwed to blocks	50-mm styrofoam panels, 75-mm air space, isolated	90-mm split-rib blocks	69
90-mm blocks 16-mm drywall screwed to blocks	125-mm air space, isolated	90-mm split-rib blocks	69
90-mm blocks 16-mm drywall screwed to blocks	50-mm styrofoam panels, 25-mm air space, structurally connected	90-mm split-rib blocks	62*
* Limited by structural transmission			

It is difficult to give reliable STC ratings for cavity block walls. Even in the laboratory, where construction is carefully controlled, it is possible to get widely differing answers. In practical installations, great care, careful supervision, and a good design are needed for two-leaf block walls to achieve their potential.

Concrete lock walls with Added Drywall

Drywall is commonly used to finish concrete block walls. It can be applied directly or supported at a distance from the block surface. One goal of the OCBA project was to study the effects of different methods of attaching drywall to a 190-mm (nom. 8") block wall. Ideally, the added drywall should not be solidly connected to the block. This prevents sound from travelling directly from the drywall to the block as vibration in solid materials. In practice, if the drywall supports are sufficiently resilient, the vibration will be attenuated. The methods used included some common techniques (13-mm resilient metal channels, 40-mm wood furring, 65-mm steel studs) and alternatives that are not in common use (50- and 75-mm Z-bars). Another furring system that could be used, but which was not tested, is 38-mm wood strapping and 13-mm resilient metal channels. This should perform about the same as 50-mm Z-bars. For each method of attachment, the wall was tested with the cavity empty and filled with glass-fibre batts. Adding absorptive material to the cavity is most beneficial when the connections are either not solid or resilient.

Figure 5 gives two results for walls with unfilled cavities. The striking feature in this plot is that at low frequencies the transmission loss has decreased relative to the unfinished block, while at high frequencies, it has increased. The drywall behaves like a drum skin resonating at a particular frequency known as the mass-air-mass resonance frequency. Around this frequency, vibrational energy will transfer from the drywall through the air in the cavity to the block more effectively than in the case of the bare block. This increases sound transmission of the wall at the important low frequencies, and can even result in a lower STC. In Figure 5, with the 75-mm Z-bars the resonance occurs at 80 Hz. For the 13-mm resilient metal channels, the resonance is less well defined (around 200 Hz). The larger the air space or the heavier the added drywall, the lower the frequency at which the mass-air-mass resonance occurs.

Adding sound-absorbing material to the cavity lowers the mass-air-mass resonance frequency. The transmission loss curve above the resonance is generally higher and the STC improves too. This is confirmed by the results in Figures 6a and 6b. As well as reducing the frequency of the mass-air-mass resonance, filling the cavity with absorptive material reduces the effects of resonances at higher frequencies and can reduce the detrimental effects of leaks around power outlets and the like. Adding sound absorbing material to the cavity is a common way to improve the sound insulation of walls or floors. Fibrous materials used for thermal insulation, such as cellulose fibre, glass-fibre or mineral wool insulation, are good materials for this purpose. Closed-cell thermal insulators, such as polystyrene, do not absorb much sound.

This resonant behaviour occurs in any type of wall constructed as layers with air spaces between them, not just in block walls. Unless measurements are made at low frequencies, however, the mass-air-mass resonance is not always visible on transmission loss plots. Many common lightweight partition designs suffer from having the mass-air-mass resonance frequency too high. The deficiencies are clear when a stereo system plays in the adjacent apartment; the melody line is barely audible, but the low-frequency bass line thumps through.

Figure 7 shows results for walls with drywall applied to one side and both sides of the blocks. The addition of the second layer further improves transmission loss at higher frequencies and further reduces it around the low-frequency mass-air-mass resonance.

Table 3.Recommended minimum cavity depths (mm) for adding drywall layers to concrete block

	Number of layers of 13-mm drywall		Number of layers of 16-mm drywall	
	1	2	1	2
no sound-absorbing material	90	45	75	40
with sound-				

absorbing material	65	30	55	30
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Table 4. STC ratings for 190-mm normal-weight block walls with different methods of mounting 16-mm drywall, with and without glass-fibre batts or other sound-absorbing material in the cavities

	No absorber		With absorber	
	One side	Both sides	One side	Both sides
Bare blocks	50			
applied directly	50	49		
40-mm wood furring	53	54	55	59
13-mm resilient channels	51	49	54	49
50-mm resilient furring	52	52	59	64
65-mm steel studs	58	57	60	72
75-mm resilient furring	57		61	

The improvement in transmission loss due to the added drywall begins two to three 1/3-octave bands above the mass-air-mass resonance. To maximize the improvement due to the airspace, the mass-air-mass resonance frequency should be as low as practical. As a general guideline, designing for a mass-air-mass resonance of 63 Hz means that the presence of the cavity improves transmission loss from 125 Hz upward, thereby ensuring an increase in STC. Table 3 gives corresponding recommended minimum cavity depths for single and double layers of drywall.

There is some evidence that lightweight, porous block walls with drywall added on furring can give better sound insulation than similar constructions with normal-weight blocks. The block porosity appears to increase the effective cavity depth, lowering the mass-air-mass resonance frequency and offsetting the lighter weight of the block. This possibility has not yet been investigated properly. There is probably an optimum range of porosity for the block.

Table 4 summarizes STC results for the 190-mm block walls measured in the OCBA series.

Prediction of STC for other block walls

The extensive series of measurements provided enough data to allow an empirical method for predicting results to be developed and verified. Briefly, the idea is that a given drywall and furring combination will have the same relative effect on any block wall that has material properties similar to the 190-mm blocks used in this study. Calculating the transmission-loss curve for another block requires only the addition of specific correction terms to the measured transmission loss for the bare block at each frequency. This technique has been applied to the 140-mm (nom. 5-1/2 ") concrete block walls shown in Figure 2; these were also measured as part of the OCBA study. Like the 190-mm block in the study, the measured transmission loss did not change when these blocks were sealed. Table 5 gives measured and predicted STC ratings for these blocks with different drywall support systems. Thus, given the transmission loss data for a block that has negligible leakage through its pores, it is possible to calculate the system performance with reasonable accuracy. To estimate the improvement in STC for different thicknesses of normal-weight block, even when the transmission loss values are not known, it can be assumed that the relative increase will be about the same as it was for the 190-mm blocks. For example, adding 16-mm drywall, glass-fibre batts and 50-mm Z-bars to both sides of the 190-mm block increased the system STC from 50 to 64. The STC for a 90-mm normal-weight block wall should also increase by about 14 points. This gives only rough estimates but might help readers to make choices.

Table 5. STC ratings for 140-mm normal-weight block walls

Description		Measured STC	Predicted STC
Side 1	Side 2		
1. bare	140-mm 75% solid blocks bare	47	
2. 65-mm glass fibre batts, 65-mm steel studs and 16-mm drywall	bare	61	60
3. as 2.	40-mm glass-fibre batts 40-mm wood strapping and 13-mm drywall	67	66
4. bare bare	as 3. paint 140-mm 100% solid blocks bare	55	54 48
5. paint	bare	50	
6. paint	40-mm glass-fibre batts 40-mm wood strapping and 13-mm drywall	58	57

It might be thought that attaching drywall directly to the surface of the block would eliminate the air gap and any mass-air-mass resonance. This is not so. There is still a small gap and, although the resonance moves to frequencies around 400 Hz, the STC can be reduced by 1 or 2 points relative to the bare block, when this method of attachment is used.

Summary

With little effort on the part of the designer, block walls provide good sound insulation. With some understanding of the acoustics, excellent walls can be constructed to meet the most demanding situations. Two-leaf block walls provide excellent transmission loss values in principle, but they must be constructed with great care. In general, it seems preferable to choose some combination of block and drywall where it is easier to ensure that the drywall is mounted in the correct, resilient fashion.

Further Reading

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